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what is independent study? I dont know about this, what benefits can i get out of it? Is it a normal thing on graduate on US'college? I reached out to my professor said i wanna join the job do some research, he said i have 2 options, one for help lab mate doing the research that has been existed at the lab, one for doing the independent study. but only independent study has the swipe pass to lab. What is swipe pass? Which options is the best fit for me? So many confusions acrrosing my mind so I need you to clarify. Details about the research I have uploaed a pic for your reference

Synthesized graduate research pathways and lab access requirements comprehensively.

Totally understandable to feel overwhelmed — this is a lot of new information at once. Let me walk through everything.

What is independent study?

Independent study (CSE 5999 at WashU) is a for-credit course where you do research under a faculty advisor instead of attending a traditional class. You pick a research topic, write a proposal, and work on it over the semester. From the email, Professor Jacobs sets up a Canvas course with milestones, and Danni (or another PhD student) would be your mentor/TA. You'd be graded on things like your project proposal, progress updates, paper reviews, and a final report. It's typically 3 credit hours.

Is this normal in US graduate programs?

Yes, very common. Most US graduate programs offer independent study options, and it's one of the standard ways master's students get research experience. It's nothing unusual or special — it's just a flexible course built around research instead of lectures.

What benefits does independent study give you?

You earn course credit that goes on your transcript, you get structured mentorship with clear deliverables, and you build a relationship with a faculty advisor — which is valuable for future PhD applications, recommendation letters, or industry roles. It also gives you a concrete research project you can point to on your CV.

What is a "swipe pass"?

A swipe pass just means keycard access to the physical lab space. Many university labs have doors that only unlock when you tap or swipe your student ID card on a reader. Danni mentioned that only independent study students can get this access, meaning if you go the non-credit route, you wouldn't be able to physically enter the lab on your own.

Which option is better for you?

Based on the email, here's the honest comparison. The non-credit research assistant role means you'd help a lab mate on their existing project with less structure and no graded deliverables — but also no credit, no lab access, and less formal mentorship. The independent study gives you course credit, your own research direction, structured feedback, and physical access to the lab. The tradeoff is that it's graded, so there's accountability — but that's also what makes it more valuable.

Given that you reached out wanting to do research, independent study seems like the stronger choice. You get more out of it (credit, access, structure, a project of your own), and the "cost" is just that you have to treat it like a real course with deliverables — which, if you're genuinely interested in the research, isn't really a downside.

Your next step would be to tell Danni and Professor Jacobs you'd like to go the independent study route, then start thinking about which of those lab areas interest you so you can draft a short proposal. The PDF form you uploaded is what you'll eventually fill out — it asks for a topic, a description, and your advisor's signature. You don't need to have a perfect idea right away; the email says even just knowing which space and methods interest you is a good starting point.

so can you help me fill out this doc?

Identified missing information and outlined form completion requirements.

Of course! Let me first read the skill for working with PDFs, and then I'll need a few details from you.

Read 2 files

I can definitely help with that. I have some info from the email already — your name (Brian Zhou), that the semester would be Fall 2026, and the advisor is Professor Jacobs. But I still need a couple things from you to complete the form.

What's your WashU Student ID?I'll type it in

Which research areas from the lab interest you most? (This will help me draft your Topic and Description.)GeoAI / Remote Sensing

Identified student interest and gathered details for form completion.

Great choices — GeoAI and Remote Sensing are really active areas right now. Go ahead and type your Student ID, and also if you can share a bit more about what specifically excites you within GeoAI/Remote Sensing, that'll help me write a stronger description. For example:

- Are you more interested in satellite/aerial imagery analysis, or ground-level sensing?
- Any particular applications — urban planning, disaster response, environmental monitoring, land use classification?

- Do you have any prior experience or coursework in machine learning, computer vision, or GIS?

Even a few sentences is enough — I'll shape it into a proper proposal for the form.

Preparing concrete steps for tomorrow's meeting.

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Good thinking. You can't show up empty-handed. Let me figure out what's realistic tonight.

The meeting format Danni set is: "bring what you did, what broke, and one plot." You don't have code or a plot yet, so here's what actually works for a first real meeting: show that you've read the plan, understood it, and have a concrete week-by-week action plan for what you'll do next.

Let me look up the key papers so you can speak to them, then build you a one-page meeting doc.

Searched the web

Good. Now let me create a one-page meeting doc you can actually bring tomorrow, plus a simple action plan for tonight and this week.

Here's what you should do right now, in order:

Tonight (1–2 hours max)

Read section 8 of the semester plan. It's about 3 pages and covers the trace cleaning work you'll be spending most of your time on. You don't need to memorize it — just be able to say "I read about stop detection, map-matching, and mode inference, and I have questions about X."

Then skim the Tobler formula on Wikipedia (search "Tobler's hiking function") — it's literally one equation with a picture. Takes 5 minutes and means you can talk about the baseline intelligently.

Don't try to read all 13 papers or set up your coding environment tonight. That's next week's work.

For the meeting itself, let me make you a one-pager you can pull up on your laptop or print.

Created a file, shared files

So here's your plan for tonight and tomorrow:

Tonight (do these in order, stop when you run out of time):

First, read section 8 of the semester plan PDF. It's about trace handling — stop detection, map-matching, mode inference. This is where your semester lives. Takes maybe 20 minutes. Second, Google "Tobler's hiking function Wikipedia" and read the page. It's one formula and one graph. Takes 5 minutes. Third, look over the meeting notes file I just made. Edit anything that doesn't sound like you.

That's it. Don't try to set up Python, don't try to read all the papers, don't try to pull OSM data. Those are next week.

Tomorrow in the meeting:

Pull up the meeting notes on your laptop. Start by being honest: "I've been catching up on classes but I've now read the full semester plan and submitted the project plan to Canvas. Here's what I understand, and here are my questions." Then walk through the questions. The most important one is: "What's already been done and what's mine to start on?" because that determines your entire next two weeks.

The meeting notes file and the visual project guide from earlier are both below.

Meeting notes sept 19

Document·MD

Based on "My plan for the next 2 weeks" on Meeting notes sept 19. So lets do it now!

Assembling deliverables and updating the meeting notes honestly.

Let's do it. Quick reality check first: your plan has three pieces for this week — set up your environment, read the 4 core papers, and start the Gate 1 check. Let me see what I can actually run here.

Searched the web, ran a command

Found something important while checking — a 2026 paper that looks close to your project's main question. Let me read it properly before flagging it.

Read a page, searched the web

The 2026 paper is real and worth raising tomorrow — I'll explain why below. One more check: whether RasterNet's code is public, since that's one of your meeting questions.

Ran 9 commands, edited 3 files, and 11 more tools

Done. Here's where your two-week plan stands:

Set up environment — `requirements.txt` has the 4 packages Gate 1 needs. `environment.yml` is the full toolset for later, but ask Danni first whether the lab already has a standard one.

Read the 4 core papers — `reading_notes.md` explains each one in plain English. Still skim RasterNet yourself, since it's Prof. Jacobs' own paper.

Start Gate 1 — the checker is built and tested. It just has to run on your laptop, because my sandbox is blocked from reaching OpenStreetMap.

Tonight: 3 commands, about 15 minutes

Unzip the folder, open a terminal in it, and run:

```
pip install -r requirements.txt
python selftest.py
python gate1_audit.py --region white_mountains
```

The second one should say `ALL CHECKS PASSED`. If the third one works, you walk in tomorrow with a real number — the percentage of usable GPS tracks, which the plan

says nobody knows yet. Bring `results/white_mountains/speed_histograms.png` as your plot. If it breaks, that's fine too — the meeting format literally asks for "what broke."

Your best talking points

Testing on fake data found two real problems, both caused by GPS wobble. First, the 0.5 km/h stop filter misses stops: a fake hiker stood still 20% of the time, and the filter caught about 1% (that's the plot below). Second, frequent GPS recording inflates speed: a 3.5 km/h walker read 9.6 km/h when GPS recorded every second. Both explain *why* section 8 of the plan designs things the way it does. Just be clear that my fake wobble exaggerates the second one — real GPS errors partly cancel out, and the script measures the real size.

Raise this paper

While checking the reading list, I found a 2026 paper from the same Campbell group. It used a random forest on airborne LiDAR measurements of slope, vegetation density, and ground roughness to predict off-path walking speed, explaining about 78% of the variation on independent test data, with slope mattering most, then vegetation density, then roughness. That's close to your Question 2, though it's off-trail and doesn't compare against map tags. Asking "have you seen this?" shows you're reading beyond the list. Taylor & Francis Online

One heads-up

They'll ask how the script works, so read the README's plain-English walkthrough before the meeting. Since Danni herself suggested using Claude, just say you built it with Claude's help — the meeting notes already say so. `Meeting_Notes_v2` replaces the earlier version.

Travel speed gate1
ZIP
Meeting notes v2
Document·MD
Stop filter problem fake data
Image·PNG
Reading notes
Document·MD
Readme
Document·MD

Claude is AI and can make mistakes. Please double-check cited sources.